Practice Exam 2 · Short Version: Solutions

50 points. Grade yourself only after a timed attempt.

Part A (6 p)

#	Answer	Concept
1	**C**	Leakage: the offer exists only after the member has started to cancel.
2	**D**	Resampling inflates probabilities (ranking fine, size wrong); p* needs calibrated sizes.
3	**A**	Lower threshold → more flagged, recall up, precision down.
4	**B**	1 − 0.95^30 ≈ 0.79 (data snooping).

5 (2 p)

False. A small p-value shows an association in past data, not the effect of changing the input; members who visit more may simply be more motivated, so only a randomized test shows what pushing visits up does.

Part B (17 p)

B1 (3 p)

```
z = −1.80 + 0.90·1 + (−0.30)·2 = −1.50
e^−1.50 = 0.223 ;  p = 0.223 / 1.223 = 0.18
```

About **18%**.

B2 (5 p)

(a) e^−0.30 ≈ **0.74**: each extra visit per month multiplies the odds of cancelling by about 0.74 (about 26% lower odds), **holding contract type fixed**. Association, not cause.

(b) Different units: contract is a 0/1 switch, visits is per single visit. Three fewer visits (+0.90) equals switching to a monthly contract, and over 0–20 visits the visits term spans 6.0 log-odds. Compare realistic changes, not raw coefficients; size is not cause.

B3 (6 p)

(a) **Not supported.** p shows association; visits likely mark motivation (confounding), and visits nudged by emails may not carry the same effect. The model predicts risk, not the effect of an email.

(b) Randomly assign at-risk members to **reminder emails vs none**; compare the **60-day cancellation rate**. Randomization makes the groups alike on motivation and everything else, so the difference is caused by the emails.

(c) Baseline balance; emails actually delivered; cancellations recorded the same way in both groups; enough members.

B4 (3 p)

(a) **Model 3:** training 0.98 vs test 0.67, a huge gap and the worst test score (60 extra variables fit noise).

(b) **Model 2:** best test AUC (0.79), small gap (0.81 → 0.79), still few inputs to explain. Check the survey measures are available for everyone before the prediction date.

Part C (15 p)

C1 (5 p)

```
Recall = 78/(78+42) = 78/120 = 0.65
Precision = 78/(78+102) = 78/180 = 0.43
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Recall 0.65: the offer reaches 65% of members who would cancel; 42 of 120 are missed. Precision 0.43: about 4 in 10 offers go to members who really were about to cancel.

C2 (5 p)

```
(a) 4,000·18 + 500·248 = 72,000 + 124,000 = 196,000
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(b) **Model at 0.10**: lowest cost, 244,000 below offering everybody and 284,000 below offering nobody (if capacity allows).

(c) "Offer nobody" is 88% accurate while catching zero cancellers; accuracy weighs a 4,000 miss like a 500 offer, so 90.2% hides that the 0.50 model still misses 84 of 120.

C3 (5 p)

(a) **0.50 (50) and 0.20 (180)** are feasible; 0.10 (350) is not.

(b) **Choose 0.20.** It is feasible and cheaper than 0.50 at both ends of the cost range (135,000 vs 175,000; 387,000 vs 679,000), so it is the best choice whatever the true miss cost.

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(c) 387,000 − 268,000 = 119,000 per 1,000 members (≈ 700 per extra contact for 170
extra)
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Part D (12 p)

Item 1 (6 p)

Claim	Verdict	Reason
1 · 89% accurate → ready	**Not supported**	Offering nobody is already 88% accurate (12% cancel); need recall, cost vs baselines, calibration.
2 · At 0.20 flags 78 of 120	**Supported**	TP = 78 at 0.20 → recall 0.65.
3 · AUC 0.81 → 81% of flagged cancel	**Not supported**	AUC measures ranking; the share of flagged who cancel is precision, 43% at 0.20.

Item 2 (6 p)

JUDGMENT (35 words): The model ranks cancellation risk well (test AUC 0.81). Within the 200-contact limit, threshold 0.20 costs 219,000 per 1,000 members versus 480,000 doing nothing. It predicts risk; it does not prove the offer keeps members.

ACTION (33 words): From Monday, the retention-team manager contacts members scored ≥0.20 (about 180 per 1,000); a random 10% get no offer, to measure its effect. Monitor monthly: 60-day cancellation rate, offer versus no-offer group.